

Fields

⊙ Class	DSCI590
---------	---------

HMDA Public LAR Data Processing Metadata

1. Study Goal

Item	Description
Dataset	Public HMDA Loan/Application Register data
Main purpose	Analyze mortgage application decisions and test whether LLM-generated decisions differ under emotional or discrimination-related appeal prompts
Target variable	<code>action_taken</code>
Main outcome	Approval vs Denial
Main modeling population	Applications with <code>action_taken</code> in <code>[1, 2, 3]</code>
Excluded outcomes	Withdrawn, incomplete, purchased loans, and preapproval-only outcomes

2. Target Variable Metadata

Field	Decision	Processing	Reason
<code>action_taken</code>	Keep	Keep only values <code>1</code> , <code>2</code> , and <code>3</code>	These represent clean final application outcomes: originated, approved but not accepted, and denied
<code>action_taken</code>	Transform	Convert to binary target	Values <code>1</code> and <code>2</code> are treated as approved; value <code>3</code> is treated as denied

Recommended target encoding:

```
df["target"] = df["action_taken"].apply(lambda x: 1 if x in [1, 2] else 0)
```

Meaning:

```
1 = Approved / originated  
0 = Denied
```

3. Sample Filtering Metadata

These fields are mainly used to define a clean study population.

Field	Decision	Processing	Reason	Values
<code>derived_loan_product_type</code>	Keep for filtering	Filter to conventional first-lien loans	This combines loan type and lien status. Using this field creates a cleaner sample of standard conventional first mortgages	- Conventional:First Lien - FHA:First LienVA:First Lien - FSA/RHS:First Lien - Conventional:Subordinate Lien - FHA:Subordinate Lien - VA:Subordinate Lien - FSA/RHS:Subordinate Lien
<code>conforming_loan_limit</code>	Keep for filtering Methodology (GitHub)	Filter to <code>C</code>	Keeps conforming loans only and removes	- C (Conforming) - NC (Nonconforming)

Field	Decision	Processing	Reason	Values
			nonconforming/jumbo loans, which may follow different underwriting standards	• U (Undetermined) • NA (Not Applicable)
business_or_commercial_purpose	Keep for filtering	Filter to 2	Removes business or commercial-purpose loans because they are different from regular consumer mortgage applications	• 1 - Primarily for a business or commercial purpose • 2 - Not primarily for a business or commercial purpose • 1111 - Exempt
initially_payable_to_institution	Keep for filtering	Filter to 1	Keeps loans initially payable to the reporting institution, so the record more directly reflects that institution's own credit decision	• 1 - Initially payable to your institution • 2 - Not initially payable to your institution • 3 - Not applicable • 1111 - Exempt
applicant_age/ageapplicant	Keep for filtering	Remove 8888 and 9999	Removes unavailable or invalid age categories	• <25 • 25-34 • 35-44 • 45-54 • 55-64 • 65-74 • >74 • 8888
occupancy_type	Keep for filtering	Filter to 1	Principal residence, second residence, and investment property can have different underwriting standards	• 1 - Principal residence 89% • 2 - Second residence 2% • 3 - Investment property 9%

Filter:

```
df = df[
    (df["derived_loan_product_type"] == "Conventional:First Lien") &
    (df["conforming_loan_limit"] == "C") &
    (df["business_or_commercial_purpose"] == 2) &
    (df["initially_payable_to_institution"] == 1) &
    (df["occupancy_type"] == 1) &
    (df["action_taken"].isin([1, 2, 3])) &
    (~df["applicant_age"].isin(["8888", "9999"]))
]
```

Check exact string first:

```
df["derived_loan_product_type"].value_counts()
```

4. Final Keep Feature Metadata

These fields are kept for modeling, prompt construction, or fairness comparison.

Field	Category	[Decision]	Value	Keeping Reason
loan_purpose	Loan feature	Keep ?	• 1 - Home purchase 45% • 2 - Home	Loan purpose affects approval behavior because home

Field	Category	[Decision]	Value	Keeping Reason
			improvement • 31 - Refinancing • 32 - Cash-out refinancing 17% • 4 - Other purpose 13% • 5 - Not applicable	purchase, refinance, cash-out refinance, and home improvement loans may have different risk patterns
loan_amount	Financial feature	Keep	numeric	Loan amount is directly related to underwriting, affordability, and risk
combined_loan_to_value_ratio/loan_to_value_ratio	Financial feature	Keep ? Missing 33%	numeric	CLTV is an important underwriting variable because it measures loan size relative to property value
debt_to_income_ratio	Financial feature	Keep ? Missing 35%	numeric	DTI is directly related to affordability and mortgage approval decisions
property_value	Financial feature	Keep ? Missing 21%	numeric	Property value affects collateral and loan risk
income	Financial feature	Keep Missing 14%	numeric	Income is directly related to borrower affordability
derived_ethnicity	Demographic/fairness feature Methodology (GitHub)	Keep	• Hispanic or Latino • Not Hispanic or LatinoJoint • Ethnicity Not Available • Free Form Text Only	Needed to test whether LLM responses differ by ethnicity-related information or appeal prompts
derived_race	Demographic/fairness feature Methodology (GitHub)	Keep	• American Indian or Alaska Native • Asian 15% • Black or African American • Native Hawaiian or Other Pacific Islander • White 44% • 2 or more minority races • Joint • Free Form Text Only • Race Not Available 33%	Needed to test whether LLM responses differ by race-related information or appeal prompts
derived_sex	Demographic/fairness feature	Keep	• Male 30% • Female 19%	Needed to test whether LLM

Field	Category	[Decision]	Value	Keeping Reason
	<u>Methodology</u> (GitHub)		- Joint 32% - Sex Not Available	responses differ by sex-related information or appeal prompts
<code>applicant_age/ageapplicant</code>	Demographic/fairness feature	Keep	<25 25-34 35-44 45-54 55-64 65-74 >74 8888	Needed to test whether LLM responses differ by age group or age-related appeal prompts

5. Final Modeling / Prompt Feature Set

After filtering, use these as the main feature set:

```
loan_purpose
loan_amount
combined_loan_to_value_ratio
debt_to_income_ratio
property_value
income
occupancy_type
derived_ethnicity
derived_race
derived_sex
applicant_age
```

Target:

```
action_taken / target
```

Important note:

Demographic variables are kept because the study is about **LLM bias and prompt response differences**. They should be described as **fairness or prompt-condition variables**, not simply as ordinary creditworthiness variables.

6. Final Eliminated Feature Metadata

A. Redundant Loan/Product Fields

Field	Decision	Elimination Reason
<code>loan_type</code>	Eliminate	Redundant because <code>derived_loan_product_type</code> already captures conventional loan status. FHA, VA, and USDA loans also follow different program rules, so the sample is filtered to conventional first-lien loans
<code>lien_status</code>	Eliminate	Redundant because <code>derived_loan_product_type</code> already captures first-lien status
<code>derived_dwelling_category</code>	Eliminate	Low variation; most records are single-family site-built properties

B. Non-Comparable Loan Product Fields

Field	Decision	Elimination Reason
<code>reverse_mortgage</code>	Eliminate	Reverse mortgages are different from standard mortgage applications

Field	Decision	Elimination Reason
		and most records are not reverse mortgages
open-end_line_of_credit	Eliminate	Open-end credit lines, such as HELOCs, are different from standard closed-end mortgages
business_or_commercial_purpose	Use for filtering, then eliminate	Used to keep only non-business-purpose loans; after filtering, it has no remaining useful variation
conforming_loan_limit	Use for filtering, then eliminate	Used to keep only conforming loans; after filtering, it has no remaining useful variation
initially_payable_to_institution	Use for filtering, then eliminate	Used to keep loans initially payable to the reporting institution; after filtering, it has no remaining useful variation
derived_loan_product_type	Use for filtering, then eliminate	Used to keep conventional first-lien loans; after filtering, it has no remaining useful variation
applicant_credit_score_type	Eliminate	This does not provide the actual credit score, but it controls for which credit scoring model was used

C. Post-Decision or Data Leakage Fields

Field	Decision	Elimination Reason
denial_reason-1	Eliminate	Directly leaks the outcome because it explains why a denied application was denied
denial_reason-2	Eliminate	High missingness and outcome leakage
denial_reason-3	Eliminate	High missingness and outcome leakage
denial_reason-4	Eliminate	High missingness and outcome leakage
purchaser_type	Eliminate	Post-origination field; it describes who purchased the loan after the loan decision
interest_rate	Eliminate	May be determined after approval/pricing and can create data leakage
rate_spread	Eliminate	Pricing-related and may be post-decision
total_loan_costs	Eliminate	Closing/pricing-related and may be post-decision
total_points_and_fees	Eliminate	Closing/pricing-related and highly missing
origination_charges	Eliminate	Closing/pricing-related and may be post-decision
discount_points	Eliminate	Pricing-related and may be post-decision
lender_credits	Eliminate	Pricing-related and may be post-decision

D. High-Missing or Low-Utility Financial Terms

Field	Decision	Elimination Reason
loan_term	Eliminate	Low diversity; most loans are 360-month loans
prepayment_penalty_term	Eliminate	Very high missingness
intro_rate_period	Eliminate	High missingness and mainly relevant to adjustable-rate products
total_points_and_fees	Eliminate	Very high missingness
multifamily_affordable_units	Eliminate	Very high missingness and not central to this study

E. Low-Variation Repayment Feature Fields

Field	Decision	Elimination Reason
hoepa_status	Eliminate	Low usefulness for this study and limited variation
negative_amortization	Eliminate	Very low variation; almost all records have no negative amortization
interest_only_payment	Eliminate	Low variation; most records have no interest-only payment
balloon_payment	Eliminate	Very low variation; most records have no balloon payment

Field	Decision	Elimination Reason
other_nonamortizing_features	Eliminate	Very low variation; almost all records have no other non-amortizing features

F. Property Fields Removed Due to Low Variation or Irrelevance

Field	Decision	Elimination Reason
construction_method	Eliminate	Low variation; most records are site-built
manufactured_home_secured_property_type	Eliminate	Mostly not applicable because the study focuses on standard site-built/conventional first-lien loans
manufactured_home_land_property_interest	Eliminate	Mostly not applicable and not central to the study
total_units	Eliminate	Low variation; most properties have one unit
multifamily_affordable_units	Eliminate	Mostly missing or not applicable

G. Redundant Detailed Demographic Fields

Field	Decision	Elimination Reason
applicant_ethnicity-1	Eliminate	Redundant because derived_ethnicity provides cleaner aggregated ethnicity
applicant_ethnicity-2 to applicant_ethnicity-5	Eliminate	High missingness and redundant with derived_ethnicity
applicant_race-1	Eliminate	Redundant because derived_race provides cleaner aggregated race
applicant_race-2 to applicant_race-5	Eliminate	High missingness and redundant with derived_race
applicant_sex	Eliminate	Redundant because derived_sex provides cleaner aggregated sex
applicant_age_above_62	Eliminate	Redundant because applicant_age already captures age group

H. Co-Applicant Fields

Field	Decision	Elimination Reason
co-applicant_ethnicity-1 to co-applicant_ethnicity-5	Eliminate	Often not applicable because many records have no co-applicant
co-applicant_race-1 to co-applicant_race-5	Eliminate	Often not applicable or highly missing
co-applicant_sex	Eliminate	Often not applicable
co-applicant_age	Eliminate	Often not applicable
co-applicant_age_above_62	Eliminate	Often not applicable and redundant if co-applicant age is excluded
co-applicant_credit_score_type	Eliminate	Often not applicable and does not provide actual credit score

I. Observation-Based Demographic Collection Fields

Field	Decision	Elimination Reason
applicant_ethnicity_observed	Eliminate	Describes how demographic information was collected, not the applicant's financial profile
co-applicant_ethnicity_observed	Eliminate	Co-applicant field and not central to the study
applicant_race_observed	Eliminate	Describes observation method only
co-applicant_race_observed	Eliminate	Co-applicant field and not central to the study
applicant_sex_observed	Eliminate	Describes observation method only
co-applicant_sex_observed	Eliminate	Co-applicant field and not central to the study

J. Application Process Fields

Field	Decision	Elimination Reason
preapproval	Eliminate	Preapproval status is not central to final approval/denial analysis
submission_of_application	Eliminate	Describes application channel/process rather than creditworthiness or prompt-bias conditions
aus-1	Eliminate	Automated underwriting system may reflect processing path but is not central to the LLM prompt-bias study
aus-2 to aus-5	Eliminate	High missingness and not central to the study

K. Record Identifier and Geography Fields

Field	Decision	Elimination Reason
activity_year	Eliminate	Not needed if using one year of data
lei	Eliminate	Identifies the reporting institution; may create institution-specific effects that are outside the study goal
state_code	Eliminate	Removed to avoid geographic proxy effects
county_code	Eliminate	Removed to avoid geographic proxy effects
census_tract	Eliminate	Removed to avoid geographic proxy effects and excessive location specificity
derived_msa-md	Eliminate	Removed to simplify the study and avoid regional market proxy effects

L. Census Tract Context Fields

Field	Decision	Elimination Reason
tract_population	Eliminate	Neighborhood-level context is outside the main individual-level LLM prompt-bias study
tract_minority_population_percent	Eliminate	Could introduce geographic/racial proxy effects and make it harder to isolate prompt effects
ffiec_msa_md_median_family_income	Eliminate	Market-level income context is outside the main study goal
tract_to_msa_income_percentage	Eliminate	Could introduce socioeconomic proxy effects
tract_owner_occupied_units	Eliminate	Neighborhood housing composition is outside the study goal
tract_one_to_four_family_homes	Eliminate	Neighborhood housing composition is outside the study goal
tract_median_age_of_housing_units	Eliminate	Neighborhood housing age is outside the study goal

7. Missing Value Handling Metadata

Field	Missing Issue	Processing Recommendation
debt_to_income_ratio	Around 35% missing	Keep; create missing indicator or treat NA / Exempt as separate category
combined_loan_to_value_ratio	Around 33% missing	Keep; create missing indicator and convert valid values to numeric
property_value	Around 21% missing	Keep; create missing indicator or impute
income	Around 14% missing	Keep; create missing indicator or impute
applicant_credit_score_type	Many Not applicable values	Keep as categorical control variable

Example missing indicators:

```

df["dti_missing"] = df["debt_to_income_ratio"].isin(["NA", "Exempt"])

df["cltv_missing"] = (
    df["combined_loan_to_value_ratio"].isna() |
    df["combined_loan_to_value_ratio"].isin(["NA", "Exempt"])
)

df["property_value_missing"] = df["property_value"].isna()

df["income_missing"] = df["income"].isna()

```

8. Final Data Processing Summary

Step	Processing Action	Purpose
1	Keep <code>action_taken</code> in [1, 2, 3]	Focus on clean approval/denial/origination outcomes
2	Filter to conventional first-lien loans	Remove FHA, VA, USDA, and subordinate-lien loans with different rules
3	Filter to conforming loans	Remove jumbo/nonconforming loans
4	Filter to non-business-purpose loans	Focus on consumer mortgage applications
5	Filter to loans initially payable to reporting institution	Keep records that better reflect the institution's original decision
6	Remove unavailable applicant age values	Avoid invalid demographic categories
7	Remove post-decision and leakage fields	Prevent outcome leakage
8	Remove redundant detailed demographic fields	Use cleaner derived demographic categories
9	Remove high-missing and low-variation fields	Improve data quality and reduce noise
10	Keep financial, property, and demographic prompt-bias fields	Support LLM decision and fairness-response analysis

9. Final Short Methodology Paragraph

I processed the HMDA Public LAR data by selecting fields that are available before or during the mortgage application decision and removing variables that could create data leakage, such as denial reasons, interest rate, rate spread, loan costs, points, and lender credits. I filtered the dataset to conforming, conventional first-lien, non-commercial loans that were initially payable to the reporting institution in order to create a more consistent study population. I removed fields with high missingness, low variation, redundant demographic detail, post-origination information, and geography-based variables that could introduce proxy effects. I kept applicant demographic variables, including race, ethnicity, sex, and age, because the purpose of this study is to examine whether LLM-generated mortgage decisions differ under demographic or discrimination-related appeal prompts.

10. References

10. References / Sources

The **Public HMDA LAR Data Fields** page was used as the main source for public LAR field names, field descriptions, and coded values such as `action_taken`, `loan_type`, `loan_purpose`, `construction_method`, `debt_to_income_ratio`, and other public loan/application register variables.

- The **2024 HMDA Filing Instructions Guide (FIG)** was used as a reference for the official HMDA filing structure, reporting context, and 2024 data collection/reporting guidance.

2024 FIG (Filing Instructions Guide) | HMDA Documentation

fig2024}

 <https://ffiec.cfpb.gov/documentation/fig/2024/overview>

- The **Appended HMDA Data Fields** documentation was used for fields that are added or derived for public HMDA use, including appended geographic, demographic, and derived public-use variables.

Appended HMDA Data Fields | HMDA Documentation

This guide explains fields that are included in public HMDA data collected in 2018 and beyond, that have been modified from the original loan record to protect privacy, or added to enhance the usefulness of the data. Derived fields used in HMDA data and publication products, such as Race, Sex, and Ethnicity, were aggregated from the source data, following a methodology developed by the CFPB for analysis and privacy protection, but are not an official and sole

 <https://ffiec.cfpb.gov/documentation/publications/general/derived-data-fields>

- The **CFPB HMDA Platform Derived Fields Categorization 2018 Onward** GitHub documentation was used to understand how derived HMDA fields are categorized from underlying reported fields, especially for derived demographic and product-related variables.

Derived Fields Categorization 2018 Onward

The HMDA Submission backend applications. Contribute to cfpb/hmda-platform development by creating an account on GitHub.

 <https://github.com/cfpb/hmda-platform/wiki/Derived-Fields-Categorization-2018-Onward>

cfpb/hmda-
platform

The HMDA Submission backend applications.

cfpb

 43  6  106  96
Contributors Used by Stars Forks

- The **HMDA Data Disclosure Policy Guidance Executive Summary** was used to justify why some HMDA fields are excluded, modified, rounded, binned, or reduced in precision in the public dataset. The guidance explains that the Bureau modifies public loan-level HMDA data to balance public disclosure benefits with applicant and borrower privacy risks.
- https://files.consumerfinance.gov/f/documents/HMDA_Data_Disclosure_Policy_Guidance.Executive_Summary.FINAL.122

Consumer Financial Protection Bureau. (2018). Executive summary of the HMDA data disclosure policy guidance.

Federal Financial Institutions Examination Council. (n.d.). Public HMDA - LAR data fields. HMDA Documentation.

Federal Financial Institutions Examination Council. (n.d.). 2024 Filing Instructions Guide: Overview. HMDA Documentation.

Federal Financial Institutions Examination Council. (n.d.). Appended HMDA data fields. HMDA Documentation.

Consumer Financial Protection Bureau. (n.d.). Derived Fields Categorization 2018 Onward. HMDA Platform GitHub Wiki.